ANDREAS LOEHR

Embedded Systems Engineer & ML Systems Developer

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andreas-loehr-486a52135 | 14 October 1997

Software Engineer with strong fundamentals in C++, embedded systems, and ML integration. Experienced building end-to-end systems from hardware interface to application layer; comfortable working at multiple levels of abstraction with attention to system architecture and reliability.

EXPERIENCE

Software Engineer - Machine Learning

SimPlan AG

Nov 2024 – Ongoing | Hanau, Germany

Integrated ML solutions into production VSM software, building ML backend pipelines and user-facing features.

- Value Stream Digitization: Improved ML backend for digitizing analog value streams. Implemented image preprocessing to handle large panorama images, refactored backend to use task queues for robustness, deployed via Docker Compose with Sentry monitoring.
- Value Stream Assistant: Built and integrated agentic AI assistant with voice input (Web Speech API) and evaluation pipeline. Iteratively tested and refined assistant capabilities.

Working Student - Software Engineering

SimPlan AG

Nov 2022 – Nov 2024 | Hanau, Germany

Implemented ML pipeline for value stream digitization in VSM software.

- Built three-step computer vision pipeline: object detection, OCR text detection, and edge detection. Custom backend API design; responsible for full data pipeline (collection, preprocessing, model training).
- Trained and deployed object and text detection models for production use.

Student Worker - Pandemic Modeling Tool

Goethe-University Frankfurt

Feb 2022 – May 2022 | Frankfurt, Germany

Translated discrete-time epidemic/pandemic modeling tool from Mathematica to Python.

Working Student - Risk Management

Deloitte Risk Advisory

June 2019 – May 2021 | Frankfurt, Germany

Developed data analysis and text mining tools for financial risk assessment.

- Built MS Access/VBA frontend for SQL Server backend (NII stress testing tool): document upload, analysis, simulation validation, and automated report generation with unit and integration tests.

EDUCATION

PhD Reinforcement Learning

Karlsruher Institut für Technologie (KIT)

Aug 2025 – Ongoing

- Part-time PhD; drafting research proposal; active research group engagement

M.Sc. Mathematics

Goethe-University Frankfurt

April 2022 – Nov 2024

Thesis: *Policy Evaluation in Distributional Reinforcment Learning* (Code on GitHub)

- Grade: 1.3
- Specialization: Probability Theory (Stochastic Analysis, Advanced Stochastics)
- Electives: Numerical Mathematics, Functional Analysis
- Minor: Computer Science (ML, Distributed Systems)

B.Sc. Mathematics

Goethe-University Frankfurt

April 2019 – Aug 2022

Thesis: *Learning Using Statistical Invariants* (Code on GitHub)

- Grade: 2.3
- Specialization: Probability Theory; Minor: Computer Science

B.Sc. Business Administration

University of Mannheim

Aug 2016 – July 2019

- Grade: 2.2
- Business fundamentals, finance, and management; concurrent with math studies

Semester Abroad

Singapore Management University

Aug 2018 – Dec 2018

- Named Entity Recognition prototype using Transformers; improved duplicate detection via regex and similarity measures; image preprocessing and OCR.

Internship - Risk Management

Deloitte Risk Advisory

January 2019 – March 2019   Frankfurt, Germany

Image preprocessing and OCR enhancement for financial documents.

- Developed image preprocessing pipeline using OpenCV to improve OCR accuracy on document scans.

PROJECTS

Real-time Embedded Sensing on ESP32 via ESP-IDF

Personal Project

2026   GitHub

Low-level embedded C++ application for sensor data collection on microcontroller with FreeRTOS-based architecture.

- Task-based concurrency model (WifiConnectTask, Sht3xTask, UpdateTask, TimeSyncTask) with queue-based sensor data flow and event group synchronization.
- Direct microcontroller hardware: I2C protocol handling, SHT31 sensor integration, NVS-based storage for credentials.
- HTTP client for pushing real-time telemetry; designed for deterministic operation on embedded platform.

ONNX Model Inference & Serving in C++

Personal Project

2025   GitHub

Prototypical C++ inference server with AMQP message routing for deploying and serving machine learning models.

- Model inference: Expose YOLOv11 object detection models via onnxruntime C++ API and AMQP message routing.
- OnnxInferLib abstraction layer for ONNX runtime inference with image preprocessing and postprocessing pipelines.
- Testing: C++ GTest-based unit tests; Python client for results validation.

- Focus: Quantitative Finance, Statistics, Programming

SKILLS

Python | C/C++ | Embedded Systems | Microcontrollers

Machine Learning | Model Inference | ONNX | PyTorch

ESP32 | FreeRTOS | ESP-IDF | CMake

FastAPI | Docker | Linux | Git | REST APIs | CI/CD | Real-time Systems

Computer Vision

LANGUAGES

German - Native | English - C1 | Spanish - B1 | Russian - A1

PUBLICATIONS

- Micha Selak et al. (2026). *Agentic Insight Generation in VSM Simulations*. arXiv: 2604.12421 `[cs.CL]`. URL: https://arxiv.org/abs/2604.12421.
- – (Dec. 2025). “LLM Assisted Value Stream Mapping”. In: *Proceedings of the 2025 Winter Simulation Conference*. WSC, pp. 2015–2026. DOI: 10.1109/WSC68292.2025.11338928.

INTERESTS

Embedded Systems, Microcontrollers, Machine Learning, Programming, Mathematics, Language Learning, Fitness & Nutrition